

Traveling wave differential protection applied to renewable energy and flexible DC connected lines

Binshu Chen
Department of Electrical Engineering
Tsinghua University
Beijing, China
cbs18@mails.tsinghua.edu.cn

Yuanyuan Dai
STATE GRID Shanghai Pudong
electric power supply company
Shanghai, China
995192303@qq.com

Xinzhou Dong
Department of Electrical Engineering
Tsinghua University
Beijing, China
xzd@tsinghua.edu.cn

Haozong Wang
Department of Electrical Engineering
Tsinghua University
Beijing, China
whz17@mails.tsinghua.edu.cn

Xinyuan Li
Department of Electrical Engineering
Tsinghua University
Beijing, China
lxy21@mails.tsinghua.edu.cn

Tianxiao Huang
STATE GRID Jibei electric power
company limited
Beijing, China
15117916439@163.com

Abstract—With the rapid consumption of non-renewable energy such as fossil energy, countries around the world are vigorously developing wind, light, geothermal and other renewable energy. Since most of the large-scale renewable energy power stations are located in remote areas, in order to better connect renewable energy to the grid, renewable energy is widely used through the flexible DC connection transmission scheme. The AC line of the renewable energy power station incorporated into the flexible DC double ended converter power supply has new fault characteristics. This paper introduces the application of traveling wave differential protection in this scenario, and uses Euclid distance to characterize the waveform similarity, and adopt normalization method to further optimize the protection performance and protection setting. Finally, the effectiveness of the protection algorithm is verified through simulation.

Keywords—Renewable energy; Traveling wave differential protection; MMC; double ended converter line.

I. INTRODUCTION

The sustainable supply of energy and environmental protection have become two key factors restricting the economic and social development of countries all over the world. China's economy will maintain a trend of rapid growth for a period of time in the future. The demand for energy is large, and the fossil energy, mainly coal and oil, is gradually drying up; At the same time, environmental pollution, climate change and other issues have become increasingly prominent. Therefore, vigorously developing new energy has become an inevitable choice to solve the contradiction between the rapid growth of energy demand, the shortage of energy supply and environmental problems[1].

Different from traditional synchronous generators, wind turbines, photovoltaic power supplies and other new energy sources are connected to the grid through power electronic devices, which are different from synchronous generators in terms of operation mechanism, grid connection topology and control mode[2]. As a result, the short-circuit current of renewable energy sources after fault shows different characteristics from synchronous generators, such as limited amplitude, non-power frequency and controlled phase angle. With the increasing proportion of renewable energy power in the power grid, its impact on the traditional protection system based on power frequency steady phasor is increasingly prominent[3].

The traditional power frequency protection is difficult to adapt to the change of power grid fault characteristics after the large-scale renewable energy power supply is connected, and the protection performance is degraded and even the false operation may occur[4]. The grid connection of renewable energy power supply is affected by the requirements of low voltage ride through capability. After the fault, the renewable energy power supply will continue to provide a certain short-circuit current. Affected by the control strategy, the fault characteristics of new energy power supply show that the steady-state power frequency component is greatly reduced and unstable during the main protection action time, and various integral and non-integral harmonics are greatly increased.

At the same time, with the development of renewable energy power generation, since there is no strong AC system to support renewable energy stations in remote areas, the island system, in which converters of flexible DC systems provide voltage and frequency reference for wind, light and other renewable energy sources, is a new form of renewable energy grid connection in the future. At present, both Zhongdu and Kangbanuoer stations of Zhangbei flexible DC grid have appeared all power electronic island systems of flexible DC converters - renewable energy stations.

Due to the addition of flexible DC converters, the protection performance of grid connected lines is further reduced, and the research on protection of some new energy transmission lines can not be applied to this.

References [5] proposes a novel principle of pilot protection based on structural similarity and square error method. The structural similarity criterion makes use of the difference of fault characteristics between renewable sources and synchronous generators to identify internal faults, while the square error auxiliary criterion is used to solve abnormal calculation of the existing similarity based protection. References [6] presents a current amplitude comparison-based pilot protection to deal with failure or reduced sensitivity of differential protection for phase failures. This method can be simply achieved in existing relays, so it is cost-effective. In addition, it only needs amplitude information, so it has a lower requirement for communication systems. References [7] proposes a novel singular value decomposition based pilot protection to solve this problem. A Hankel matrix is established for the short-circuit current at each end of the AC line. The Hankel matrix rapidly detects the singularity of the signal and its derivative, and the SVD method uses a short

time window to extract the waveform features. This approach enables the detection of the fault features before MMC blocking in the worst-case fault scenario and prevents the incorrect fault identification caused by the minor amplitude of fault current provided by the converters. Reference [8] proposes a pilot protection method based on Spearman's rank correlation coefficient in this article. When the system operates normally or an external fault occurs, a traversing current flows through the transmission line, so the current waveforms on both sides are the opposite. When an internal fault occurs, there are large differences in transient current waveforms on both sides. Spearman's rank correlation coefficient can be used to measure the degree of correlation between the waveforms on both sides to distinguish between faults inside and outside the zone. References [9~12] analyzes the shortcomings of traditional distance protection in renewable energy stations, and give several improved distance protection principles. This belongs to the improvement research of backup protection.

In this paper, Section II introduces the principle of traveling wave differential protection applicable used to double ended converter lines, and Section III proposes a waveform similarity measurement method based on Euclid distance on the basis of the original traveling wave differential protection principle, and sets another traveling wave differential protection criterion. The criterion has better sensitivity and can be set without system parameters. The simulation experiment in Section IV verifies the protection principle. Section V gives the conclusion.

II. INTRODUCTION OF TRAVELING WAVE DIFFERENTIAL PROTECTION PRINCIPLE

A. General Solution of Wave Equation - Traveling Wave

Transmission lines can be represented by a distributed parameter line model with uniform parameters, that is, series resistance, series inductance, parallel conductance and parallel capacitance have distributed characteristics. Figure 1 shows the equivalent circuit of a single wire distributed parameter line model with local dx length. The current reference direction is as shown in the figure. R_0 , L_0 , G_0 and C_0 are the resistance, inductance, conductance and capacitance of the line per unit length, respectively. u and i are the voltage and current on the transmission line, both functions of distance and time.

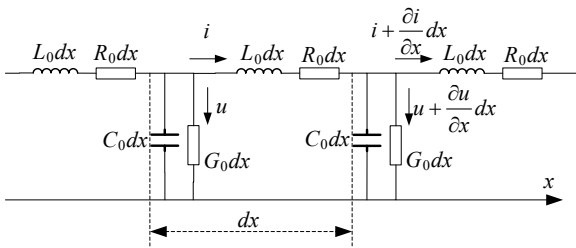


Fig. 1. Local equivalent circuit of single wire distributed parameter circuit model

According to the local equivalent circuit, the loop voltage equation and node current equation can be listed as shown in equation (1), which is the so-called transmission line equation.

$$\begin{cases} -\frac{\partial u(x,t)}{\partial x} = R_0 i(x,t) + L_0 \frac{\partial i(x,t)}{\partial t} \\ \frac{\partial i(x,t)}{\partial x} = G_0 u(x,t) + C_0 \frac{\partial u(x,t)}{\partial t} \end{cases} \quad (1)$$

By solving this equation in frequency domain, (2) is get:

$$\begin{cases} U(x, j\omega) = A_1(j\omega)e^{-\gamma x} + A_2(j\omega)e^{\gamma x} = U_f(x, j\omega) + U_r(x, j\omega) \\ I(x, j\omega) = \frac{A_1(j\omega)e^{-\gamma x}}{Z_c} + \frac{A_2(j\omega)e^{\gamma x}}{-Z_c} = I_f(x, j\omega) + I_r(x, j\omega) \end{cases} \quad (2)$$

Where, $A_1(j\omega)$ and $A_2(j\omega)$ is the undetermined coefficient, Z_c is the wave impedance, as shown in Formula (3)[14].

$$Z_c = \sqrt{\frac{Z_0}{Y_0}} = \sqrt{\frac{R_0 + j\omega L_0}{G_0 + j\omega C_0}} \quad (3)$$

According to the general solution formula (2) of the wave equation, the voltage and current at any point on the line are composed of two parts, namely, forward traveling wave and reverse traveling wave in the frequency domain. If the distributed resistance R_0 and the distributed conductance G_0 are not considered, voltage and current can be expressed as (4):

$$\begin{cases} U(x, j\omega) = A_1(j\omega)e^{-j\omega x/v} + A_2(j\omega)e^{j\omega x/v} \\ I(x, j\omega) = \frac{A_1(j\omega)e^{-j\omega x/v}}{Z_c} + \frac{A_2(j\omega)e^{j\omega x/v}}{-Z_c} \end{cases} \quad (4)$$

$$v = \frac{1}{\sqrt{L_0 C_0}} \quad (5)$$

Therefore, in the time domain, the voltage and current can be expressed as (6).

$$\begin{cases} u(x, t) = a_1\left(t - \frac{x}{v}\right) + a_2\left(t + \frac{x}{v}\right) = u_f(x, t) + u_r(x, t) \\ i(x, t) = \frac{a_1\left(t - \frac{x}{v}\right)}{Z_c} + \frac{a_2\left(t + \frac{x}{v}\right)}{-Z_c} = i_f(x, t) + i_r(x, t) \end{cases} \quad (6)$$

It can be seen from the general solution that the voltage and current can be decomposed into traveling wave forms moving along two directions respectively.

The assumption of lossless lines is often used in the analysis because of the obvious traveling wave motion of lossless lines and the small loss of transmission lines.

B. Traveling Wave Differential Protection Principle

The principle of traveling wave differential protection is illustrated in Figure 2, and x represents the reference positive direction.

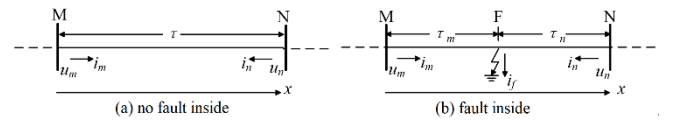


Fig. 2. fault current of MCC

In Fig. 2, the line is assumed to be a single-phase lossless line. Traveling wave differential current can be expressed as (7).

$$\begin{cases} i_{x1}(t) = 2(i_{mr}(t) - i_{nr}(t - \tau)) \\ i_{x2}(t) = 2(i_{nf}(t) - i_{mf}(t - \tau)) \end{cases} \quad (7)$$

Where, $i_{x1}(t)$ and $i_{x2}(t)$ are traveling wave differential currents at M and N terminals respectively; τ is the time when traveling wave propagates from M end to N end (or from N end to M end); $i_{mr}(t)$, $i_{mf}(t)$ and $i_{nr}(t)$, $i_{nf}(t)$ are the current reverse wave and current forward wave at M and N terminals respectively. They can be calculated through equation (8) by using voltage and current.

$$\begin{cases} i_{mf}(t) = \frac{1}{2} \left(\frac{u_m(t)}{Z_c} + i_m(t) \right) \\ i_{nf}(t) = \frac{1}{2} \left(\frac{u_n(t)}{Z_c} - i_n(t) \right) \\ i_{mr}(t) = \frac{1}{2} \left(-\frac{u_m(t)}{Z_c} + i_m(t) \right) \\ i_{nr}(t) = \frac{1}{2} \left(-\frac{u_n(t)}{Z_c} - i_n(t) \right) \end{cases} \quad (8)$$

When there is no fault inside,

$$\begin{cases} i_{x1}(t) = 0 \\ i_{x2}(t) = 0 \end{cases} \quad (9)$$

$$\begin{cases} i_{mr}(t) = i_{nr}(t - \tau) \\ i_{nf}(t) = i_{mf}(t - \tau) \end{cases} \quad (10)$$

According to the invariance of traveling wave transmission, after time delay τ , the forward wave is equal to the reverse wave, and traveling wave differential current is zero.

When there is a fault inside,

$$\begin{cases} i_{x1}(t) = i_{f1}(t - \tau_m) \\ i_{x2}(t) = i_{f2}(t - \tau_n) \end{cases} \quad (11)$$

Where $i_{f1}(t)$, $i_{f2}(t)$ are two components of fault branch current and their sum is the fault branch current, τ_m and τ_n is the time of traveling wave propagation in MF segment and FN segment respectively. Due to the existence of the fault, the traveling wave cannot be transmitted to the opposite end unchanged, there is obvious traveling wave differential current.

The principle of traveling wave differential protection adopts distributed parameter circuit, which can naturally avoid the influence of capacitance current. At the same time, the principle uses the instantaneous value of voltage and current, which is not affected by the harmonic component in the fault electrical quantity. This principle can effectively improve the performance of line protection with double ended converter.

The setting of protection criteria is shown in Formula (12),

$$\begin{cases} \sum |i_{mr}(t) - i_{nr}(t - \tau)| > I_{set} \\ \sum |i_{mr}(t) - i_{nr}(t - \tau)| > K_{set} \end{cases} \quad (12)$$

The first one is the amplitude criterion, and only when the absolute integral value of traveling wave differential current reaches a certain value in the protection time window can it act. The second is the ratio criterion, which introduces the braking current. In case of external fault or no fault, the braking current is equal to twice the line current, and the differential current is equal to zero. In case of internal fault, the differential current increases significantly, and the ratio increases sharply.

III. TRAVELING WAVE DIFFERENTIAL PROTECTION SCHEME BASED ON EUCLID DISTANCE

The protection criterion directly uses the difference value of traveling wave as the action quantity. Although it is completely correct under ideal conditions, in practical applications, because of the existence of phase error and amplitude error of two traveling waves, the simple difference value cannot effectively represent this difference. A parameter

used to measure waveform similarity needs to be applied. At the same time, for the setting values of the braking criterion and amplitude criterion, the original scheme did not provide a simple and feasible setting method, which needs to rely on simulation and empirical values.

Euclidean distance is an effective method to calculate the space distance of two groups of space vectors $h = \{h_1, h_2, \dots, h_n\}$ and $g = \{g_1, g_2, \dots, g_n\}$. The formula is [15],

$$q = \sqrt{\sum_{i=1}^n (h_i - g_i)^2} \quad (13)$$

Through this formula, the similarity of two time series can be characterized. As a simple example, in case of external fault, current traveling wave on both sides which pass through τ is completely consistent, and the calculation result of q is 0.

The Euclid distance is used to describe the traveling wave differential value. In the formula, it is the square sum of the waveform difference. Compared with the direct waveform difference, the points with large waveform difference account for a larger proportion in the result, which can amplify the waveform difference. The measure of waveform similarity using Euclid distance can better reflect the waveform difference, and the protection has higher sensitivity.

In order to facilitate tuning, Euclidean distance is normalized, so that the similarity of traveling wave waveforms on both sides can be compared uniformly in all scenarios. The protection criteria can be de parameterized and not affected by system parameters.

$$q_1 = \frac{\sqrt{\sum_{i=1}^n (h_i - g_i)^2}}{\min(\sqrt{\sum_{i=1}^n h_i^2}, \sqrt{\sum_{i=1}^n g_i^2})} \quad (14)$$

Considering that if there is a zero sequence in h and g , that is, in the field measurement, the electric current wave on one side is always zero, the division form is written as the multiplication form.

$$\sqrt{\sum_{i=1}^n (h_i - g_i)^2} = q_1 * \min(\sqrt{\sum_{i=1}^n h_i^2}, \sqrt{\sum_{i=1}^n g_i^2}) \quad (15)$$

When there is no fault or external fault, q is close to 0 and q_1 is close to 0; In case of internal fault, there is obvious difference between current traveling waves on both sides. The magnitude of $\sqrt{\sum_{i=1}^n (h_i - g_i)^2}$ and $\min(\sqrt{\sum_{i=1}^n h_i^2}, \sqrt{\sum_{i=1}^n g_i^2})$ is equal or even the former is much larger than the latter.

Set protection criteria,

$$q_1 > q_{set} \quad (16)$$

Expressed in multiplication form,

$$\sqrt{\sum_{i=1}^n (h_i - g_i)^2} > q_{set} * \min(\sqrt{\sum_{i=1}^n h_i^2}, \sqrt{\sum_{i=1}^n g_i^2}) \quad (17)$$

The line voltage and current are periodic sinusoids in steady state, and the main frequency component in fault is the grid frequency. In order to make the calculation result of Euclid distance more stable, the protection time window should be an integral multiple of 0.5 cycle. Here, we use 50Hz grid frequency, and the protection time window is 20ms. The sampling frequency is 10kHz.

The setting of q_{set} should be greater than that of the most serious fault outside. q_{set} considers the maximum error when there is no fault and external fault. The error mainly consists of two parts: 1. amplitude error. Suppose that the amplitude of current traveling wave on one side is small, and the amplitude of electric current traveling wave on the other side is 1.1 times; 2. Time synchronization error. In consideration of 10kHz sampling, the time synchronization error is taken as 0.05ms half of the sampling interval, and the delay error of voltage and current transformers and the synchronization error of protection devices are taken as 0.5ms,, then the total time synchronization error is taken as 0.55ms, corresponding to the phase error of 9.9°.

It is assumed that the current traveling wave in case of fault is approximately sine wave,

$$g(t) = A_f \sin(\omega t) \quad (18)$$

$$h(t) = 1.1A_f \sin(\omega(t - 0.00055)) \quad (19)$$

Where, A_f is the amplitude of current traveling wave in case of fault, ω is 100 π .

$$q_1 = \frac{\sqrt{\sum_{i=1}^{200} (1.1A_f \sin(\omega(0.1i - 0.00055)) - A_f \sin(0.1\omega i))^2}}{\sqrt{\sum_{i=1}^{200} (A_f \sin(0.1\omega i))^2}} \quad (20)$$

$$= \frac{2.0678}{10} = 0.20678$$

The setting of the criterion retains a certain margin, and the q_{set} is set to 0.4. It can be found that after normalization, the setting of protection is not affected by system operation mode and power supply type, which is more convenient for setting.

Through the analysis in this section, the traveling wave differential protection principle based on Euclid distance has higher sensitivity, and the setting is not affected by system parameters, so it is easier to conduct protection setting.

IV. SIMULATION

The simulation system is built as shown in Fig. 3. The research line is the line where the fault identification is located in the figure. One end of the bus is the collection bus of the new energy station, and the other end is the converter station at the flexible and direct transmission end. The voltage level is 220kV. The line is 30km long.

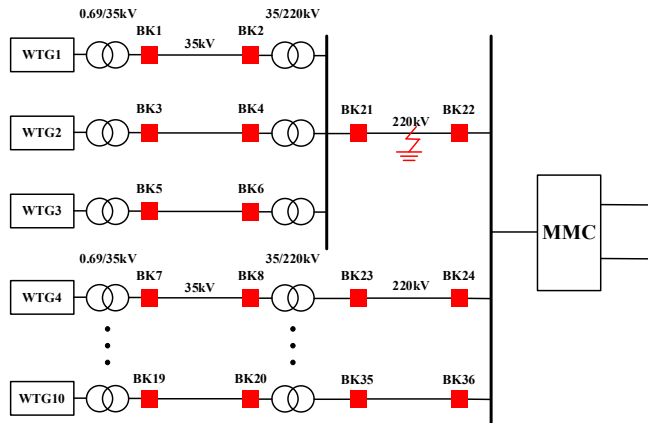


Fig. 3. simulation system

Set faults of different fault types and locations to verify the correctness of protection actions.

The protection test results are shown in Table 1.

Table 1 System capacity

Fault location	fault type	Protection action
5km from the flexible DC converter station	AG-0 Ω	action
	AG-10 Ω	action
	AG-127 Ω	action
	ABG	action
	BC	action
	ABCG	action
15km from the flexible DC converter station	AG-0 Ω	action
	AG-10 Ω	action
	AG-127 Ω	action
	ABG	action
	BC	action
	ABCG	action
25km from the flexible DC converter station	AG-0 Ω	action
	AG-10 Ω	action
	AG-127 Ω	action
	ABG	action
	BC	action
	ABCG	action
Renewable energy bus	ABCG	no action
Flexible DC converter station bus	ABCG	no action

The protection principle proposed in this paper is that the fault in the area acts correctly, and the fault outside the area does not act reliably. It can operate reliably under the condition of high resistance in the area and has high sensitivity.

V. CONCLUSION

With the massive development and utilization of renewable energy, large-scale grid connection of renewable energy has become the trend of power grid development. Due to the lack of strong AC system support for new energy stations in remote areas, the island access system, which provides voltage and frequency reference for wind, light and other new energy sources by the converter of flexible and straight systems, is a new form of renewable energy grid connection in the future.

The traditional power frequency protection is difficult to adapt to the large-scale new energy power supply at one end, and the characteristics of the flexible DC converter station at one end change, so the protection performance drops and even the false operation may occur.

This paper proposes a traveling wave differential protection principle based on Euclid distance. The traveling wave differential protection principle based on Euclid distance has higher sensitivity, and the setting is not affected by system parameters, so it is easier to conduct protection setting. Finally, the effectiveness of the protection principle is verified by simulation.

ACKNOWLEDGMENT

Funded by the State Grid Corporation of China Headquarters Science and Technology Project(5100-202114563A-0-5-SF).

REFERENCES

- [1] K. Jia, Z. Yang, Y. Fang, T. Bi and M. Sumner, "Influence of Inverter-Interfaced Renewable Energy Generators on Directional Relay and an

- Improved Scheme," in IEEE Transactions on Power Electronics, vol. 34, no. 12, pp. 11843-11855, Dec. 2019, doi: 10.1109/TPEL.2019.2904715.
- [2] D. Zhang, J. Ma, S. Wang, Y. Ji and C. Qu, "Analysis on the Characteristic of the Inrush Transient Current of Brushless Doubly Fed Induction Generator," in IEEE Transactions on Energy Conversion, doi: 10.1109/TEC.2022.3163853.
 - [3] S. J. Salon, T. Pham, C. Kaminski and L. T. Ergene, "Short circuit analysis of large doubly-fed induction generators," 2013 International Electric Machines & Drives Conference, 2013, pp. 146-151, doi: 10.1109/IEMDC.2013.6556246.
 - [4] A. Banaïmoqadam, A. Hooshyar and M. A. Azzouz, "A Control-Based Solution for Distance Protection of Lines Connected to Converter-Interfaced Sources During Asymmetrical Faults," in IEEE Transactions on Power Delivery, vol. 35, no. 3, pp. 1455-1466, June 2020, doi: 10.1109/TPWRD.2019.2946757.
 - [5] L. Zheng, K. Jia, T. Bi, Z. Yang and Y. Fang, "A Novel Structural Similarity Based Pilot Protection for Renewable Power Transmission Line," in IEEE Transactions on Power Delivery, vol. 35, no. 6, pp. 2672-2681, Dec. 2020, doi: 10.1109/TPWRD.2020.2973505.
 - [6] K. Jia et al., "Pilot Protection Based on Amplitude Comparison for Renewable Power Teed Lines," in CSEE Journal of Power and Energy Systems, vol. 8, no. 6, pp. 1519-1529, November 2022, doi: 10.17775/CSEEJPES.2019.02220.
 - [7] L. Zheng, K. Jia, B. Yang, T. Bi and Q. Yang, "Singular Value Decomposition Based Pilot Protection for Transmission Lines With Converters on Both Ends," in IEEE Transactions on Power Delivery, vol. 37, no. 4, pp. 2728-2737, Aug. 2022, doi: 10.1109/TPWRD.2021.3115117.
 - [8] K. Jia, Z. Yang, L. Zheng, Z. Zhu and T. Bi, "Spearman Correlation-Based Pilot Protection for Transmission Line Connected to PMSGs and DFIGs," in IEEE Transactions on Industrial Informatics, vol. 17, no. 7, pp. 4532-4544, July 2021, doi: 10.1109/TII.2020.3018499.
 - [9] C. Chao, X. Zheng, Y. Weng, Y. Liu, P. Gao and N. Tai, "Adaptive Distance Protection Based on the Analytical Model of Additional Impedance for Inverter-Interfaced Renewable Power Plants During Asymmetrical Faults," in IEEE Transactions on Power Delivery, vol. 37, no. 5, pp. 3823-3834, Oct. 2022, doi: 10.1109/TPWRD.2021.3138128.
 - [10] S. Paladhi and A. K. Pradhan, "Adaptive Distance Protection for Lines Connecting Converter-Interfaced Renewable Plants," in IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 9, no. 6, pp. 7088-7098, Dec. 2021, doi: 10.1109/JESTPE.2020.3000276.
 - [11] Z. Yang, K. Jia, Y. Fang, Z. Zhu, B. Yang and T. Bi, "High-Frequency Fault Component-Based Distance Protection for Large Renewable Power Plants," in IEEE Transactions on Power Electronics, vol. 35, no. 10, pp. 10352-10362, Oct. 2020, doi: 10.1109/TPEL.2020.2978266.
 - [12] Y. Fang, K. Jia, Z. Yang, Y. Li and T. Bi, "Impact of Inverter-Interfaced Renewable Energy Generators on Distance Protection and an Improved Scheme," in IEEE Transactions on Industrial Electronics, vol. 66, no. 9, pp. 7078-7088, Sept. 2019, doi: 10.1109/TIE.2018.2873521.
 - [13] L. Tang, X. Dong, S. Shi and B. Wang, "Travelling wave differential protection based on equivalent travelling wave," 13th International Conference on Development in Power System Protection 2016 (DPSP), 2016, pp. 1-6, doi: 10.1049/cp.2016.0119.
 - [14] A. Lei, X. Dong, "Wide-frequency and Short-time Window Based Traveling Wave Differential Protection Method for EHV/UHV Transmission Lines," in Proceedings of the CSEE, vol. 39, no. 23, pp. 6826-6835, Dec. 2019, doi:10.13334/j.0258-8013.pcsee.181793.
 - [15] Z. Yang, W. Liao, H. Wang, C. L. Bak and Z. Chen, "Improved Euclidean Distance Based Pilot Protection for Lines With Renewable Energy Sources," in IEEE Transactions on Industrial Informatics, vol. 18, no. 12, pp. 8551-8562, Dec. 2022, doi: 10.1109/TII.2022.3148318.